PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

(only local-network and networkwalks site results are considered)

Pentester Name	MUHAMMAD YOUSUF
Date	11 November, 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM2 (Google Hack DataBase) W2-PM4 (theHarvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. My own local LAN Network 2. Networkwalks (secured written permission already)
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only.

Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools. This report presents the practical activities performed like network scanning, website reconnaissance, and information gathering. Various cybersecurity tools were used to identify network devices, analyze domain information, examine web technologies, and collect publicly available data.

All commands were run in Kali Linux and on a Windows PC. Every step below includes the methodology used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

Objectives

- Identify active network devices, IP addresses, and MAC addresses using Zenmap.
- Perform domain and website reconnaissance using specialized tools.
- Analyze HTTP headers and DNS records.
- Identify web technologies and Web Application Firewalls (WAFs).
- Gather publicly available information using theHarvester.
- Explore GHDB for identifying potentially exposed information.
- Document technical procedures, results, and observations.

Tools Used

1. **Kali Linux** – Penetration testing and cybersecurity operating system.
2. **Windows** – for running VM and local scanning etc.
3. **Zenmap** – Network scanning and host discovery.
4. **cURL** – HTTP requests and header analysis.
5. **DNSRecon** – DNS enumeration and information gathering.
6. **Wafw00f** – Web Application Firewall detection.
7. **WHOIS** – Domain registration information lookup.
8. **WhatWeb** – Website technology identification.
9. **theHarvester** – OSINT and public information gathering.
10. **Google Hacking Database (GHDB)** – Identifying publicly indexed information using specialized search queries.

Results

Local Network Scanning (Zenmap)

Successfully scanned the local network and discovered **5 active hosts**.

- **Local IP:** `192.168.100.86`
- **Subnet Mask:** `255.255.255.0`
- **Network Range:** `192.168.100.0/24`
- **Total Live Hosts:** 5

Discovered Devices:

IP Address	MAC Address
192.168.100.1	70:79:90:8B:6F:0B
192.168.100.2	4A:E5:2E:AA:2C:47
192.168.100.3	A4:6C:F1:ED:A2:EA
192.168.100.32	80:9D:65:6C:34:DC
192.168.100.86	08:00:27:05:02:CF

Website Reconnaissance (NetworkWalks.com)

The reconnaissance activities revealed the following information:

Domain Information (WHOIS)

- Registrar: GoDaddy.com, LLC
- Creation Date: 16 April 2015 | Expiry Date: 16 April 2027
- Nameservers: `NS35.DOMAINCONTROL.COM`, `NS36.DOMAINCONTROL.COM`
- Registrant Privacy: Domains By Proxy, LLC
- DNSSEC: Unsigned

IP & Server Information

- Website IP: `192.232.216.135`
- Web Server: Apache
- Hosting Provider Infrastructure: HostGator
- Detected Server Location: United States

DNS & Email Records (DNSRecon)

- A Record: `192.232.216.135`
- NS Records: `ns6135.hostgator.com`, `ns6136.hostgator.com`
- SOA: `ns6135.hostgator.com`
- MX Record: `mail.networkwalks.com` → `192.232.216.135`
- SPF Record: `v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all`

- Nameserver IPs: `50.87.144.87`, `192.232.216.131`
- DNS Software: BIND `9.16.23-RH`
- SRV Records: 8 autodiscover records pointing to `cpanelemaildiscovery.cpanel.net` on port 443.

Website Technologies (WhatWeb)

- CMS: WordPress 7.1.4
- Web Server: Apache
- Framework: Bootstrap
- JavaScript Library: jQuery 3.7.1
- Plugin: WordPress Download Manager 3.3.76
- Google Tag Manager detected
- Contact Email: `info@networkwalks.com`

HTTP Analysis (cURL)

- HTTP Status: `301 Moved Permanently`
- Redirect: HTTP to HTTPS
- Cookie: `__wpdm_client` (HttpOnly)
- Cache-Control: `max-age=3600`
- Referrer Policy: `no-referrer-when-downgrade`

WAF Detection (Wafw00f)

- Detected Firewall: ModSecurity (SpiderLabs)
- Detection Requests: 2

Overall Result: Successfully identified local network devices and gathered domain registration, DNS, email infrastructure, server, website technology, HTTP, and firewall information through reconnaissance techniques.

Risk Observations

- **DNSSEC:** DNSSEC is reported as unsigned, meaning DNS responses do not benefit from DNSSEC validation.
- **Server Information Exposure:** Web server and DNS software details were identifiable through reconnaissance.
- **Technology Disclosure:** WordPress, plugins, and JavaScript technologies were publicly detectable.
- **Email Security:** SPF records were identified; further verification of email security configurations may be beneficial.
- **WAF Protection:** ModSecurity was detected, indicating the presence of a web application firewall.

Recommendations

- Consider implementing DNSSEC to strengthen DNS integrity.
- Minimize unnecessary exposure of server and software version information.
- Keep WordPress, plugins, and website components updated.
- Review SPF, DKIM, and DMARC configurations for email security.
- Regularly review and update WAF rules and security configurations.

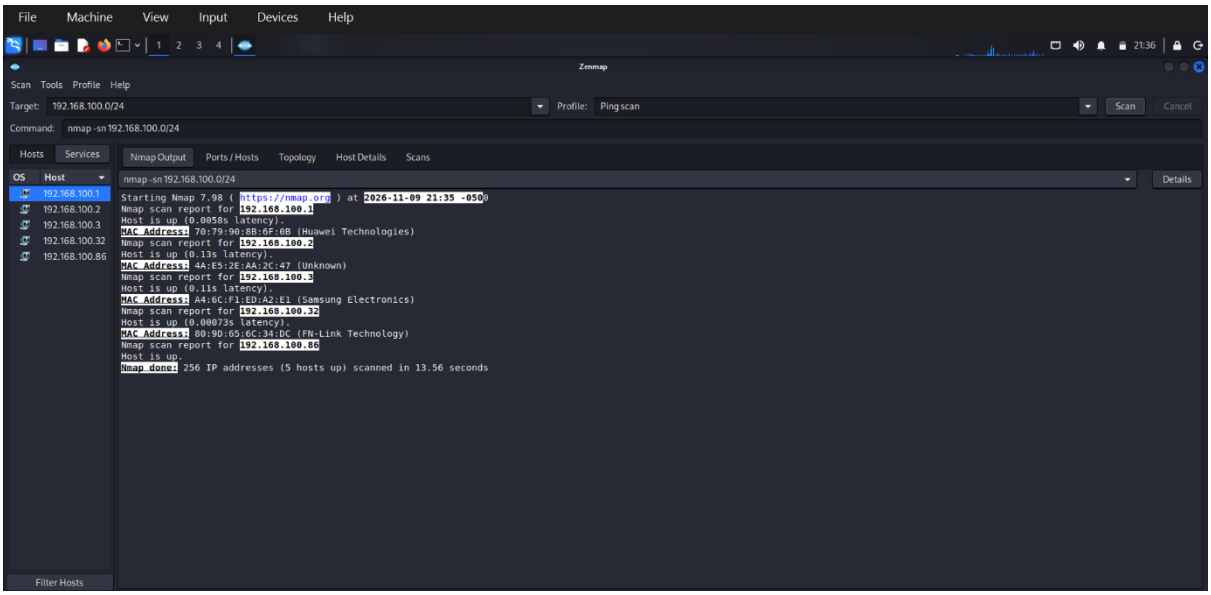
Conclusion

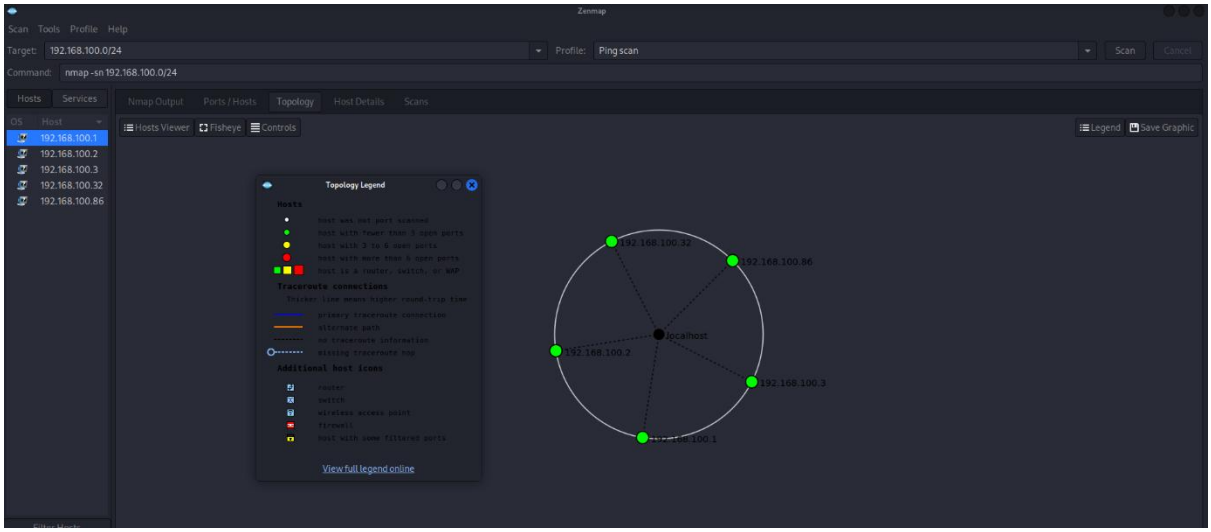
In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

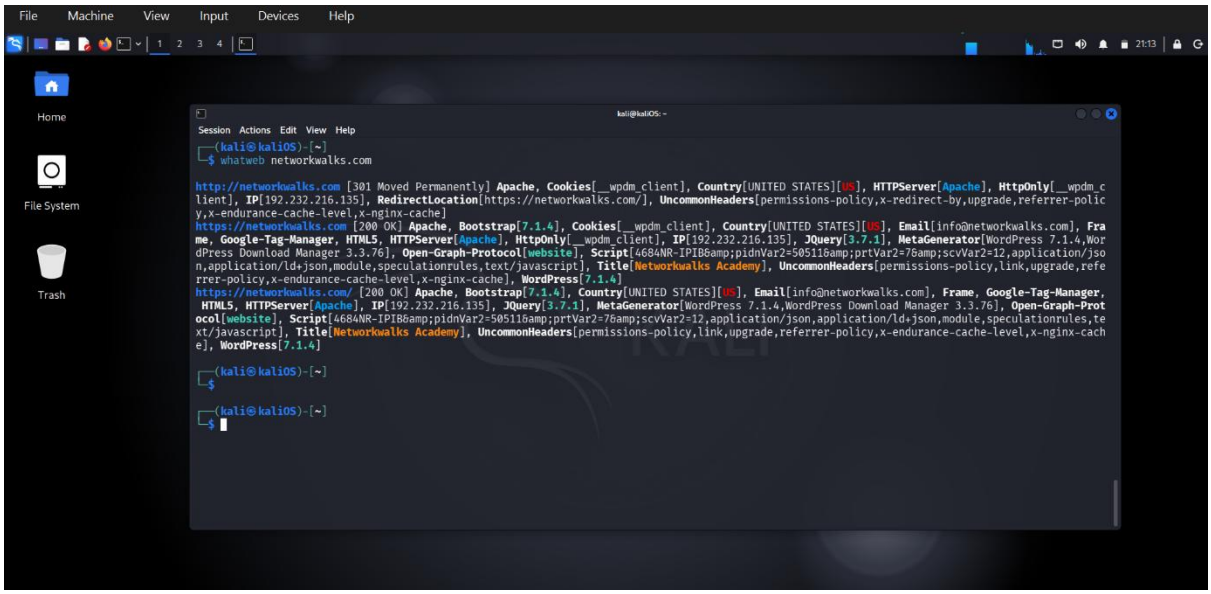
In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

Evidences Collected




Zenmap
Scan Tools Profile Help
Target: 192.168.100.0/24
Profile: Ping scan
Scan
Cancel
Command: nmap -sn 192.168.100.0/24
Hosts
Services
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
OS Host
192.168.100.1
192.168.100.2
192.168.100.3
192.168.100.32
192.168.100.86
Hosts Viewer
Fisheye
Controls
Legend
Save Graphic
Topology Legend
Hosts
Traceroute connections
Additional host icons
View full legend online
192.168.100.32
192.168.100.86
localhost
192.168.100.2
192.168.100.3
192.168.100.1


File Machine View Input Devices Help
Home
File System
Trash
kali@kali0S: ~
Session Actions Edit View Help
(kali㉿kali0S)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=505118&prtVar2=76&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=505118&prtVar2=76&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
(kali㉿kali0S)-[~]
$
(kali㉿kali0S)-[~]
$


File Machine View Input Devices Help
Home
File System
Trash
kali@kali0S: ~
Session Actions Edit View Help
(kali㉿kali0S)-[~]
$ whois networkwalk.com
Domain Name: NETWORKWALK.COM
Registry Domain ID: 1920256084_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2026-04-13T02:24:12Z
Creation Date: 2015-04-16T06:46:05Z
Registry Expiry Date: 2027-04-16T06:46:05Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS35.DOMAINCONTROL.COM
Name Server: NS36.DOMAINCONTROL.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-11-10T02:01:30Z <<<
For more information on Whois status codes, please visit https://icann.org/epp
NOTICE: The expiration date displayed in this record is the date the
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